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Estimation of crop angle of inclination for lodged wheat using multi-sensor SAR data

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ABSTRACT

Lodging - the bending of crop stems - reduces the quantity and quality of cereal crop yields. Early quantification of crop lodging is important to prevent further losses and to facilitate harvesting operations. Crop angle of inclination (CAI) is a quantitative measure of the lodging stage and a component of lodging severity/score. CAI is an important structural parameter for lodged crops and very few studies have investigated its estimation using satellite-based remote sensing. In this study, the performance of Sentinel-1 and multi-incidence angle (FQ8-27° and FQ21-41°) RADARSAT-2 data were investigated for estimating CAI. Temporal crop biophysical/structural parameters (CAI and crop height) and meteorological data (rainfall and wind speed) were collected throughout May 1-June 30, 2018 in a very large commercial farm located in Jolanda di Savoia, Ferrara, Italy. Field data were grouped into different crop lodging stages (non-lodged/healthy (H), moderate lodging (ML), severe lodging (SL) and very severe lodging (VSL)) based on CAI. Quantitative relationships were established between field-measured CAI values and the RS-derived metrics for Sentinel-1 and RADARSAT-2 timeseries using support vector regression (SVR) models. The RADARSAT-2 FQ8 model performed most robustly with a R^2_{CV} (cross-validated R^2) of 0.87 and a $RMSE_{CV}$ (cross-validated RMSE) of 8.89° while the performance of the Sentinel-1 and RADARSAT-2 FQ21 models were comparable with an $RMSE_{CV}$ of 11.35° and 11.63° respectively. Low incidence angle data were particularly sensitive to high CAI values (VSL) while high incidence angle data were useful for predicting lower CAI (ML and SL). While the RADARSAT-2 FQ-8 model outperformed the other two, the Sentinel-1 model still explained 78% of the CAI variability in the study site, which is important in the context of operational crop lodging stage assessment. This is the first study to demonstrate the utility of SAR remote sensing data for estimating CAI as a measure of the lodging stage and a component of lodging severity.

1. Introduction

Crop lodging is the permanent bending of plant stems from the vertical or displacement of the root anchorage, causing destruction of canopy structure, degradation of grain quality, slowed harvest, increased drying costs and severe yield reductions (up to 80%) (Berry et al., 2004; Pinthus, 1974). Crop lodging is caused by the complex interaction between crop's genetic, environmental and management factors, making every lodging event unique with different onset, duration and intensity (Nafziger et al., 1986; Zhu et al., 2016; Piñera-Chavez et al., 2016). The assessment of lodging extent and severity is challenging due to these complex interactions, the heterogeneous and random distribution of lodging within a field, the absence of a standard scale to represent it and the lack of data related to lodging (e.g. year wise statistics of percentage area lodged for different crops at local, regional or global scale). Plant physiologists and agronomists have studied crop lodging intensively (Baker et al., 1998; Easson et al., 1993; van Delden et al., 2010; Wu and Ma, 2016; Lang et al., 2012) but the scope is mainly limited to breeding trials (i.e. producing lodging resistant varieties) and agronomic management (i.e. agronomic practices that can reduce lodging risk). The results of these studies, as reviewed by Chauhan et al. (2019) show that three main factors control the severity of lodging and govern the extent of yield loss - the time of its occurrence (crop growth stage), the spatial extent of lodging, and the lodging angle or crop angle of inclination (CAI). CAI is defined as the angle made by the crop stem with respect to the vertical. During the process of lodging, a crop undergoes a series of stages (hereafter referred to as lodging stages) starting with a slight lean from the vertical

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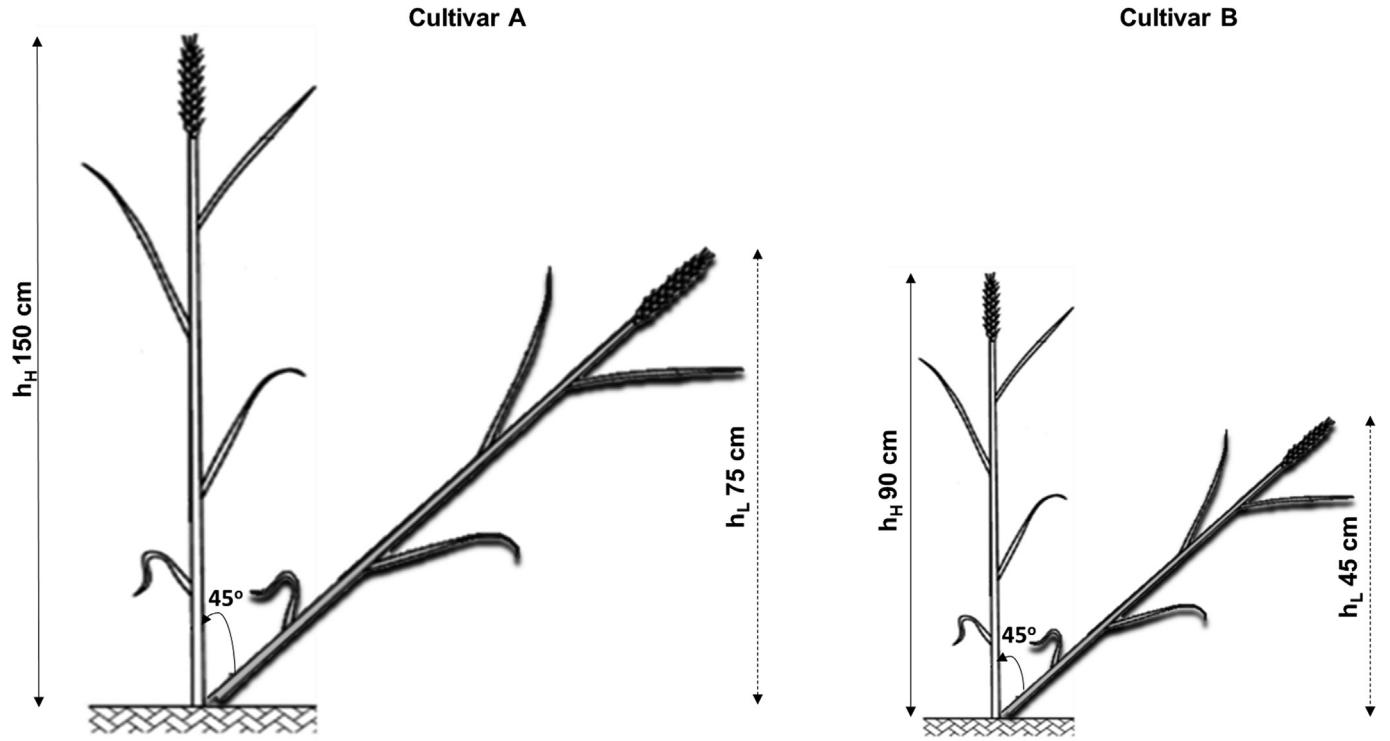


Fig. 1. An example of the change in crop height and crop angle of inclination in the event of lodging for (a) cultivar A and (b) cultivar B, at the same growth stage. h_H is the height before lodging, while h_L is the height of the lodged crop (figures not to scale) *(1.5-column-fitting)*.

(CAI ~ 0°) and ending with the crop lying close to horizontal (CAI ~ 90°). CAI is, therefore, an important metric to describe the physical structure of a lodged crop or the lodging stage (an index based on CAI-moderate, severe or very severe).

Accurate estimation of CAI can contribute to the estimation of crop yield loss. For instance, Fischer and Stapper (1987) demonstrated that the yield losses in wheat incurred at a CAI of 80° were almost between 2 and 4 times that those at 45°. CAI, when combined with lodging area percentage can also help assign a lodging score (a lodging severity index combining CAI and lodged area) to a crop. Quantitative estimates of CAI can thus be beneficial to farmers (to plan remedial actions, for example, to minimize the harvesting losses and settle compensation disputes) and insurance loss adjusters (to get an estimate of the extent of damage). The conventional methods to measure CAI and assess lodging stages rely on intensive manual ground measurements and visual ratings. As with most ground measurement strategies, such methods are time consuming and severely restrict their application for monitoring large areas repeatedly. In addition, high spatial variability associated with lodging makes it challenging to capture this variability via ground measurements.

Remotely sensed imagery offers an efficient way to obtain timely information on the temporal trends and spatial distribution of lodging. This is possible with frequent acquisitions of high-resolution imagery during the crop growth period or at least during the critical growth stages. High quality optical remote sensing data may not be consistently available due to cloud, rain and haze interference. In this context, synthetic aperture radar (SAR) sensors offer a clear advantage since microwaves are less affected by the atmosphere and can image the Earth's surface successfully irrespective of weather conditions. Also, the unique sensitivity of microwave scattering to crop structure, has led to many studies on the use of SAR for crop monitoring (Cookmartin et al., 2000; Lopez-Sanchez et al., 2011; Nelson et al., 2014a; Steele-Dunne et al., 2017; Yuzugullu et al., 2017). However, a recent review by Chauhan et al. (2019) revealed that only eight peer-reviewed articles published between 1984 and 2018 have focused on the use of SAR data for lodging assessment, four of which utilized satellite-based remote sensing (Chen et al., 2016; Han et al., 2017; Yang et al., 2015; Zhao et al., 2017). Most of these studies investigated the behaviour of remote sensing signals with respect to the condition of the lodged crop (mainly to perform detection) and there is limited knowledge on the use of SAR data for the quantitative assessment of crop parameters associated with lodging.

Multi-configuration (multi-incidence angle and multi-polarization) data from RADARSAT-2 has made it possible to address crop-lodging assessment in more effective ways. For instance, a correlation analysis between RADARSAT-2 derived parameters and lodging in wheat by Yang et al. (2015) showed that polarimetric ratios, especially those based on odd/double scattering and span (i.e. Odd/Span, Double/Span) can distinguish healthy and lodged wheat fields while a single channel backscattering coefficient (σ^o) such as σ^o_{HH}, σ^o_{HV} or σ^o_{VV} cannot separate the two classes efficiently. However, given the sensitivity of the polarization of microwaves to crop structure and dielectric properties (Srivastava et al., 2009), the capability of PolSAR in crop lodging assessment is highly dependent on the crop type (Mascolo et al., 2016). Zhao et al. (2017) performed a study to test this hypothesis. The authors examined σ^o and polarimetric features derived from RADARSAT-2 data to detect lodging in wheat and canola fields. Some features such as σ^o_{VV}, σ^o_{HH}, the depolarisation degree, and the circular-pol correlation coefficient were highly sensitive to lodged wheat, unlike canola, whose canopy structure is highly random. In another study, Chen et al. (2016) showed that σ^o_{HV}, as well as features such as those referring to the double scattering and volume scattering and the T22 and T33 matrix elements derived from RADARSAT-2 data, are capable of detecting lodging in sugarcane. Unfortunately, the above studies do not provide quantitative estimates of lodging-related crop parameters.

To date, quantitative assessments of lodging (e.g. lodging stage) using remote sensing are also hampered by the lack of dense time-series data at suitable spatial resolution. An analysis of dense time series data over the crop-growing season has the potential to estimate the date of lodging incidence and can enable close to real-time lodging monitoring.

A new era started with the launch of Sentinel-1 (hereafter abbreviated as S-1), offering a unique opportunity to monitor lodging systematically. In a recent study, Han et al. (2017) used S-1 data to develop an index to classify lodging into different stages (mild, moderate and severe) using crop height as a proxy variable. Although their model demonstrated some potential to identify and classify lodging stages, it was tested over a single growth stage (tassel) and did not exploit dense time-series data. Furthermore, as the crop inclines at a certain angle during lodging, the effective crop height, which is defined as a distance between the stem base and the head of the crop, also changes (van Delden et al., 2010). Also, the use of crop height for the interpretation of lodging stages is highly dependent on the crop variety and the growth stage. This is shown in Fig. 1. It is highly probable that at a specific growth stage, the normal crop height (h_H = 150 cm) of wheat cultivar A can be higher than that of wheat cultivar B (h_H = 90 cm). In the event of lodging, the lodged height (h_L) will, therefore, be different in both cases (say at 45°, h_L = 75 cm and h_L = 45 cm respectively). This variation makes crop height a poor indicator of lodging since there is too much of "natural" variation to use it to discriminate between lodged vs non-lodged targets without additional information. CAI, on the other hand, is independent of crop variety and the growth stage.

Estimation of crop biophysical properties from active remotely sensed data has been investigated using several modelling approaches. These models have evolved from regression and machine learning approaches such as multiple linear regression, neural networks and decision trees (Kumar et al., 2018; Prasad et al., 2012) to sophisticated radiative transfer models (Blaes et al., 2006). While these process-based models are theoretically sound in interpreting SAR data (Erten et al., 2016), the inversion of crop variables is still challenging due to inherent complexity, limited operational usage and intensive data requirement (Wang et al., 2009). Among the machine learning approaches, support vector machine regression (SVM-R or SVR) has shown excellent generalization capabilities (Blanzieri and Melgani, 2008; Tuia et al., 2011).

To the best of our knowledge, there is no research on the use of remote sensing for estimating CAI of lodged wheat. Therefore, the main objective of this study is to compare the performance of S-1 and multi-incidence angle RADARSAT-2 data for estimating CAI. The CAI estimation has been achieved by implementing SVR as a regression tool with different sets of predictor variables derived from the two SAR sensors.

2. Materials and methods

2.1. Study area

The study was carried out in the Bonifiche Ferraresi farm (Fig. 2a), situated in Jolanda di Savoia (central coordinates 44°52′59″N, 11°58′48″E), a commune in the province of Ferrara, Italy. Bonifiche Ferraresi is Italy's largest farm holding with over 6500 ha of land spread across the municipalities of Jolanda di Savoia, Arborea, Mirabello and Santa Caterina. More than 60% of the land is located in Jolanda, covering approximately 3,850 ha. The study region is mainly covered by arable land. The main crops are durum wheat (*Triticum durum*), soft wheat (*Triticum aestivum*), rice (*Oryza sativa*), corn (*Zea mays*), barley (*Hordeum vulgare*), soybean (*Glycine max*) and potatoes (*Solanum tuberosum*), among several other horticulture and medicinal plants. The climate in the region is warm and temperate. The annual mean precipitation is 691 mm, and the annual mean temperature is 13.6 °C (Fig. 2b). The soil textures are mainly clayey and silty.

2.2. In situ measurements

The fieldwork was conducted on durum and soft wheat cultivars, which were sown between Oct 21-Nov 4, 2017 in 26 fields covering approximately 600 ha. The size of the wheat fields in the farm varied between 2.38 and 84.86 ha. Winter wheat is dormant in the first few months after sowing due to low temperatures (from October to Feb). It is not until spring (from March onwards in the study area) that wheat breaks its dormancy and resumes vegetative growth.

The ground and satellite image data were collected synchronously over the period of May 1-June 30, 2018 (until harvest). However, the fields were frequently inspected from March 2018 onwards to detect the first instance of lodging (beginning of May). A stratified random sampling procedure was performed, and 76 sample plots were identified (Fig. 2a right) using six information strata (elevation, seed density, soil type, soil pH, crop variety and sowing date). Care was taken to ensure that the samples were collected from various stratum and that the entire geographical area was well represented while also considering the spatial distribution of the points. A 60 m × 60 m plot size was chosen in this study since larger plots considerably reduce the negative impact of the potential edge effects on the stability and magnitude of remote sensing-derived metrics (Frazer et al., 2011) and exhibit less between-plot variance (Zeide, 1980). Furthermore, to capture the structural variability and heterogeneity of CAI within the plots, three subplots of 2 m × 2 m were sampled in fully healthy plots (0° < CAI < 5°) while the number of subplots was increased to eight in lodged plots (6° < CAI < 90°) and the subplot CAI values were averaged. All the plots were revisited twice. However, since the first evidence of lodging was recorded during the flowering stage (on 1st May 2018 during the 2nd round of sampling), only 118 samples were considered from the observation period covering three growth stages–flowering, milk/dough development and ripening.

For each subplot, CAI was measured indirectly using a plumb bob, an inch tape and trigonometric calculations. The string of the plumb bob was suspended from the top of the plant head until the tip touched the ground. The heavyweight at the bottom ensured accurate measurement of the vertical height from the point of suspension. The total length of the suspended string (h_s) and the height of the plumb bob (h_w) were then used to calculate the vertical height (h_v) (Fig. 3). For lodged plants, the slant height (h_{sl}) was measured with an inch tape. CAI (ɵ) from the vertical was then derived for each subplot based on the measurements shown in Fig. 3a and equation (1). Subsequently, the lodging stages were defined based on CAI (ɵ) from the vertical (Fig. 3b) to carry out an exploratory discriminative analysis. Plots with an average CAI of 1-5° were categorized as healthy (H) plots while those with the average CAI ranging between 6 and 90° were classified into moderate lodging (ML), severe lodging (SL) and very severe lodging (VSL) stages (Fig. 3b). In addition, crop height of the plots was also measured using an inch tape.

We also took replicate measurements of soil moisture in each plot using a Stevens Hydra Probe. The crop growth stages were quantitatively measured using the BBCH scale (Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie) (Bleiholder et al., 2001). In each plot, the point of plant failure (stem or root lodging), amount of nitrogen applied and other structural/morphological parameters (such as leaf area index (LAI), biomass, etc.) were recorded to facilitate interpretation of the results. Daily cumulated precipitation (mm) and daily average wind speed (10 m from the ground) were continually recorded through a local automatic weather station (44°51′22.9″N, 11°57′51.0″E). The summary statistics of important field measurements are presented in Table 1. The variation of crop height (for healthy plots) at the flowering stage is plotted in Fig. 4 for different varieties. We can see in Fig. 4 that while the crop height varied from 60 cm to 133 cm, the CAI remained stable at 5°, which indicates that CAI is a better proxy variable for lodging stage assessment than crop height. The methodological flowchart of the study is presented in Fig. 5.

2.3. Remote sensing data

Single look complex (SLC) RADARSAT-2 fine Quad-pol (FQ) and SLC/GRD (ground range detected) S-1 images were acquired for the analysis. The S-1 and RADARSAT-2 images were selected synchronous

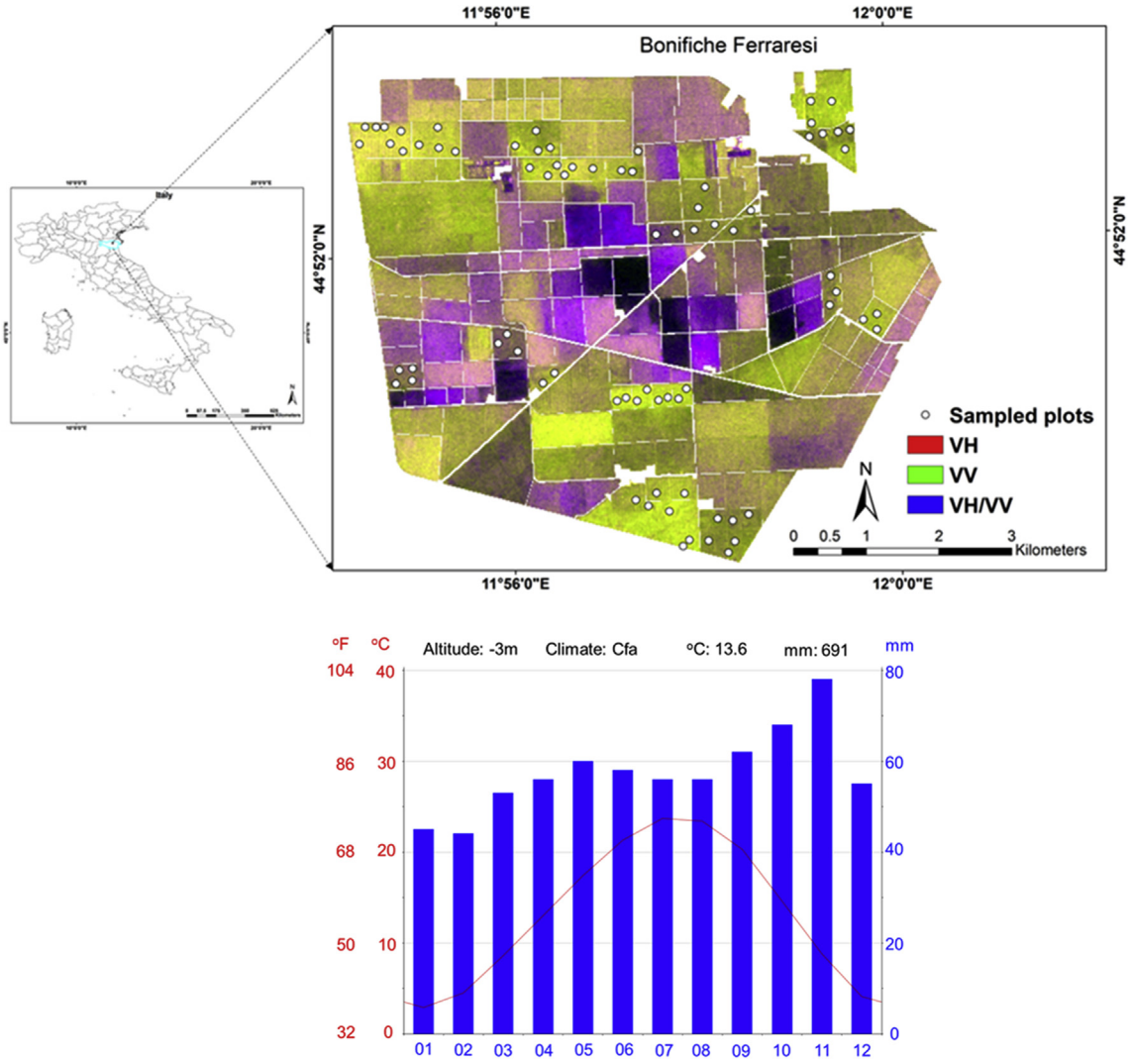


Fig. 2. (a) The study area (left) and location of the sampled plots overlaid on the S-1 RGB composite (right) acquired on 6 June 2018 and (b) Temporal distribution of precipitation throughout the year (Source: Climate-Data.org, 2015) *(1-column fitting)*.

to the dates of the field data acquisition. A set of five multi-incidence angle RADARSAT-2 images acquired over the study area were procured by the Canadian Space Agency through the SOAR (Science and Operational Applications Research) program. Two beam modes (FQ8 and FQ21) were selected to study the effect of incidence angle on the estimation of CAI and to increase the temporal data availability. In addition to RADARSAT-2 data, eleven S-1 images (incidence angle 30-46°) were acquired in Interferometric Wide-swath (IW) mode between 1 May 2018, and 30 June 2018 (Fig. 6). This mode provides dual-polarization (VV and VH) images. In this study, we used both Level-1 GRD and SLC products. The GRD products consist of focused SAR data that have already been detected, multi-looked and projected to ground-range using the WGS84 Earth ellipsoid model. SLC products, on the other hand, consist of focused SAR data and are provided in slant-range geometry. Table 2 shows some key characteristics of the acquired S-1 and RADARSAT-2 images.

2.4. Remote sensing data pre-processing

2.4.1. RADARSAT-2

The linear backscatter processing of RADARSAT-2 was performed in SARscape 5.5 while the polarimetric parameters were extracted in SNAP 6.0 software. To eliminate the orbital error in RADARSAT-2 images, definitive orbit files downloaded from the FTP repository of MDA were applied while importing the SLC products. The images were subset and co-registered using a high resolution (10 m) digital elevation model (DEM) provided by Tarquini et al. (2007). The targets under observation can have any translational and rotational motion relative to the data collection platform. Uncompensated translational motion can result in target signatures being degraded by large frequency phase errors (Werness et al., 1990). For rotating targets too, it is crucial to obtain a satisfactorily focused target image whose dimensions in azimuth are known (Werness et al., 1990). The spatial registration and resampling of the image pixels compensate for the relative translational shift, rotational and scale differences (Farghaly et al., 2019). The De Grandi spatio-temporal filter was then used to remove speckle from the images. The filter works in a combined time-space domain and preserves the temporal signature at the best (De Grandi et al., 1997). The approach outlined in Nelson et al. (2014b) was used for further processing to get normalized sigma nought or σ^o values (in dB). The normalization of backscattering coefficient helps in compensating for the range variations and is intended to empirically correct the dependency of the backscatter signal on the incidence angle. The polarimetric indices, polarimetric decomposition parameters, covariance (C3) and coherency matrix (T3) elements were also extracted from each RADARSAT-2 image (Table 3). Forty-two metrics were generated from each

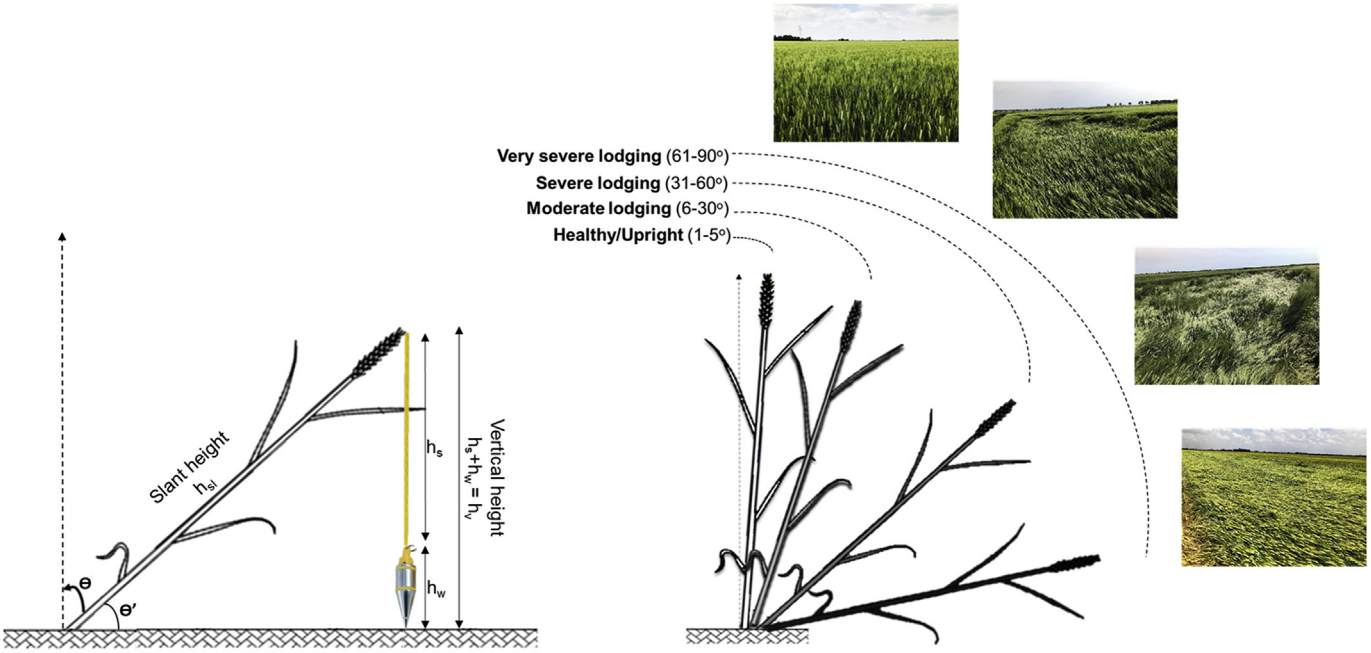


Fig. 3. (a) Measurement of crop angle of inclination and (b) Illustration of different lodging stages. *(2-column fitting)*.

$$\theta(degree) = 90^0 - \sin^{-1}\frac{h_v}{h_{sl}} \quad (1)$$

Table 1
Summary statistics of CAI and crop height for healthy and lodged plots ($n = 118$). Samples were collected throughout the flowering to ripening growth stage. *IQR* is the interquartile range, and *COV* is the coefficient of variation expressed in percentage.

Data	Median	IQR	Min	Max	Std. Dev.	COV (%)
CAI(°)	24.60	55.70	3.00	79.50	28.80	89.70
Crop height(cm)	52.22	43.25	15.00	159.30	26.92	49.50

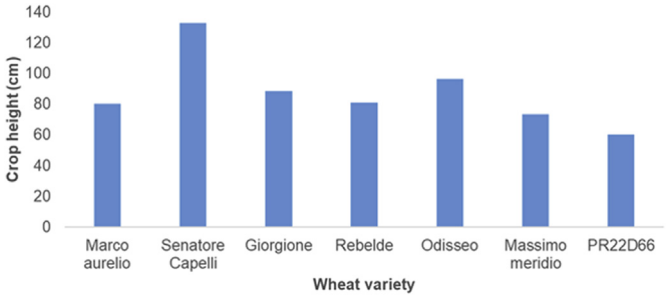


Fig. 4. Variation of crop height for the healthy plots at the flowering stage. (*1-column fitting*).

beam mode and are listed in Table 3.

2.4.2. Sentinel-1

The processing of S-1 data was performed in SARscape 5.5. The methodology of extracting linear σ^o was similar to that explained in Nelson et al. (2014b). The images were subset, co-registered, and a De Grandi multi-temporal filter was used to remove speckle from the images. For the generation of geocoded coherence maps, the coherence change detection (CCD) processing chain of SARscape was utilized. Coherence is a function of additive noise, systemic spatial de-correlation and scene de-correlation that occurs between two acquisition dates. It is sensitive to changes in either phase or amplitude of an image pixel. For instance, dielectric (wet vs dry soil) or backscattering properties both due to natural processes (e.g. crop growth) and due to abrupt changes (e.g. harvest or destruction of a crop morphology during lodging) can result in coherence loss. The interferometric coherence (γ), which ranges from 0 to 1 (1 being the perfect coherence), refers to the amplitude of the complex correlation coefficient between the two SAR images (s_1 and s_2) and is formulated as follows:

$$\gamma = \frac{|\langle s_1 s_2^* \rangle|}{\sqrt{(\langle s_1 s_1^* \rangle \langle s_2 s_2^* \rangle)}} \quad (2)$$

where s^* is the complex conjugate of s; and $\diamond$ is the ensemble average (Touzi et al., 1999). The CCD processing chain imports the SLC data, applies the orbit file correction and then subsets the data before estimating a geocoded coherence image. We used the same high-resolution DEM (10 m) to geocode and correct for topographic variations. The coherence was calculated for every adjacent image pair from an Ascending pass (e.g. date-1 and date-2, date-2 and date-3, …date-n-1 and date-n) to achieve the lowest possible temporal baseline (six days). It was assumed that there was no change in the crop lodging condition during the six-day interval (i.e. H remained H, ML remained ML and so on). Thus, five metrics (σ^o_{VV}, σ^o_{VH}, $\sigma^o_{VH/VV}$, Coherence VH, Coherence VV) were generated from S-1 data.

2.5. Exploratory statistical data analysis (ESDA)

Given the limited knowledge of the spatiotemporal dynamics of lodging, a preliminary exploratory statistical data analysis (ESDA) was carried out in MATLAB 2018b. Firstly, the behaviour of backscatter (and coherence from S-1 data) metrics derived from both satellites were analyzed and interpreted using *in situ* CAI measurements. The analysis was performed for different lodging stages (H, ML, SL and VSL).

Pearson correlation coefficients (r) were calculated between CAI and the 47 metrics (42 from RADARSAT-2 and five from S-1 data) (refer to Table 5 and Table 6). Normality was checked with the Shapiro-Wilk test while Levene's test was used to check the homogeneity of variances between different lodging stages. The Kruskal Wallis rank sum test

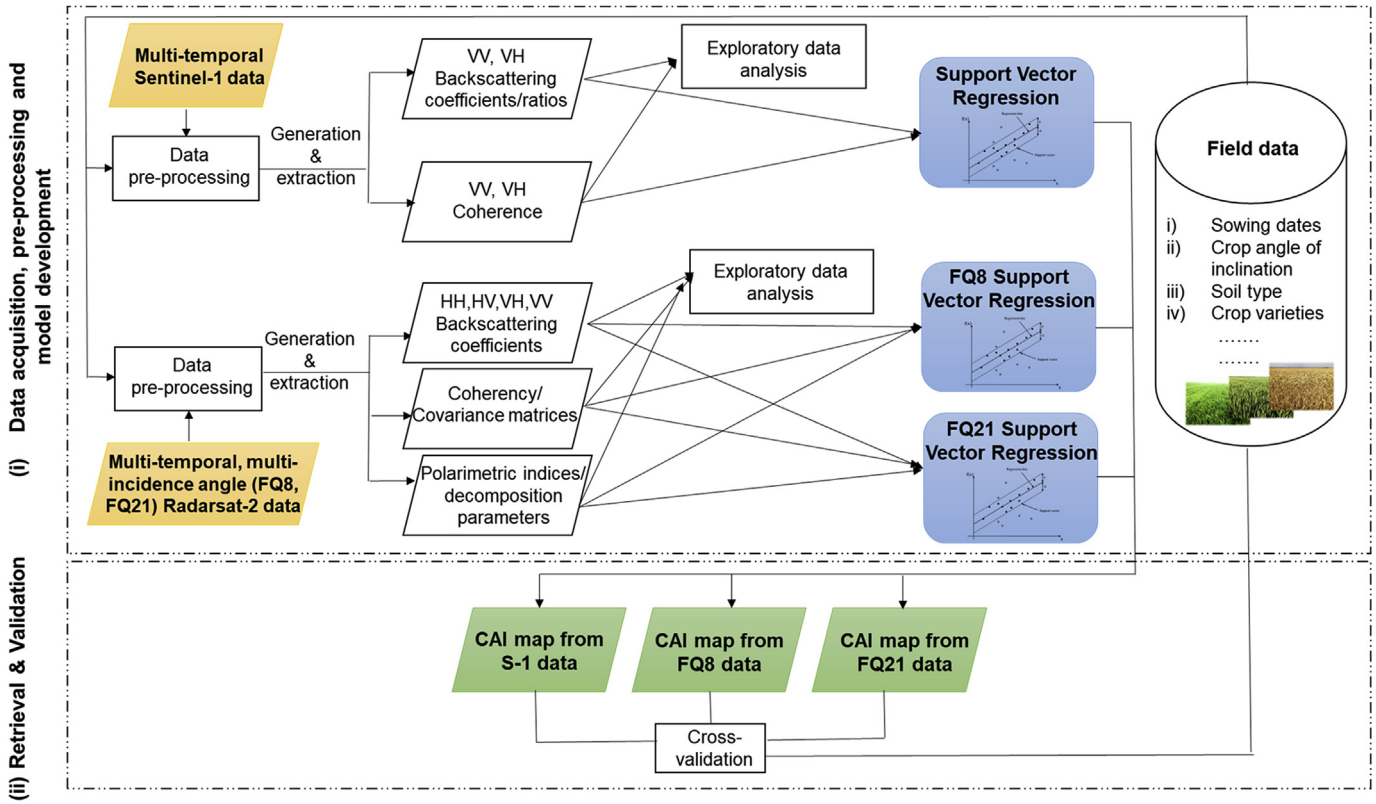


Fig. 5. Methodological flowchart of the study. (*2-column fitting*).

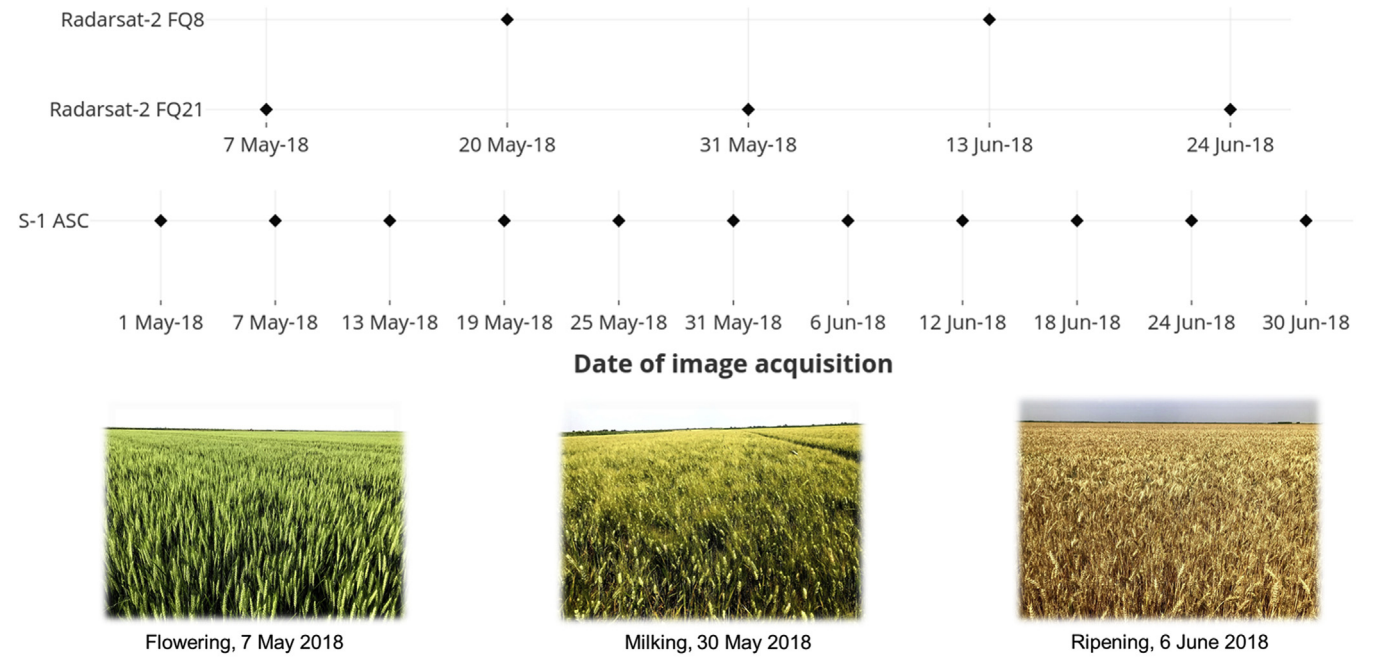


Fig. 6. Acquisition dates of the remote sensing data during the 2018 wheat growing season. (*2-column fitting*).

(Kruskal and Wallis, 1952) was used to assess the statistical differences of the sample means among the groups. Subsequently, a post hoc Tukey's Honest Significant Difference (HSD) test was also carried out to find significant pairwise differences between the categorical variables.

2.6. Support vector machine regression analysis

SVR is a non-parametric machine-learning tool that relies on kernel functions which project the input data into a new (higher dimensional) hyperspace where complex non-linear trends can be represented in a simple manner (Brereton and Lloyd, 2010; Williams, 2011). SVR aims to build an optimal hyperplane in the new hyperspace in a way that it fits the data with minimal error and complexity of the modelling function. The main advantages of SVR are (i) its ability to generalize from limited training data, (ii) convexity of the cost function allowing it to identify the optimal solution consistently, thus making it resilient to

Table 2
The image acquisition parameters for S-1 and RADARSAT-2 data. Note that the incidence angle is based on the location of the study site within the swath.

Sensor	Beam mode/Product type/ Polarization	Incidence angle range (deg.)	Pixel spacing in Range (m) × Azimuth (m)	Pixel resolution in Range (m) × Azimuth (m)	Nominal swath width (km)	Orbit direction
RADARSAT-2	SLC Fine-Quad Pol. FQ8 HH, HV, VH, VV	26.9–28.7	4.78 × 4.78	10 × 10	25 × 25	Asc
	SLC Fine-Quad Pol. FQ21 HH, HV, VH, VV	40.2–41.6	4.73 × 5.12	7 × 7	25 × 25	Des
S-1A/B	IW-GRD VV, VH	39.7–40.4	10 × 10	15 × 15	250 × 250	Asc
	IW-SLC VV, VH	39.7–40.4	2.3 × 14.1	15 × 15	250 × 250	Asc

being trapped in local minima, (iii) it is independent of the statistical distribution of the data, (iv) does not have the problem of local optimal and multicollinearity and (v) it minimizes the risk of overfitting (Bhatia and Yu-Wei, 2017).

In this study, we implemented three epsilon-SVR (ε-SVR) models to estimate CAI using S-1 ($n = 118$), RADARSAT-2 FQ8 ($n = 57$) and RADARSAT-2 FQ21 ($n = 61$) derived metrics. The models were trained and cross-validated using the partial least square (PLS) toolbox v8.7 from Eigenvector Research, Inc., with Multivariate Image Analysis (MIA) toolbox v3.0 add-on in MATLAB 2018b (Wise et al., 2006). A venetian blinds cross-validation procedure with 10 data splits was used to preserve the class proportion in each cross-validation group (Allison et al., 2009). The toolbox was operated in the default mode where it utilizes a grid search and cross-validation to select the optimal SVR parameters (such as cost, epsilon and gamma) and then build a model using those values.

The type of kernel function also governs the performance of SVR. The purpose of using a kernel function is to transform linearly inseparable data to linearly separable one by applying them on each data instance. The original (non-linear) observations are then mapped into higher-dimensional space (in which they become separable). This transformation is applied to the data before the model is trained. A Gaussian or Radial Basis Function (RBF) was used since it is known to produce better results, than other functions such as sigmoid or linear, and has fewer parameters to tune (Zuo and Carranza, 2011). The model performance was evaluated using the coefficient of determination (R^2_{Cal}, R^2_{CV}) as a measure of goodness-of-fit (between predicted and measured CAI), and root-mean-square error ($RMSE_C$, $RMSE_{CV}$). After cross-validating the performance of the models, they were applied on two S-1 (31 May and 6 June), one FQ8 (13 June) and one FQ21 (31 May) satellite images covering the whole study area to produce CAI maps. The four images with the least temporal gap were selected to enable comparative analysis.

3. Results

The first instances of lodging were observed in the fields when the crop was in the flowering stage and lodging became more severe as the crop approached maturity. During this period, the CAI varied significantly from 3° in the healthy plots to a maximum of 79.5° in very severely lodged plots with a COV of 8.21% (Table 1). Since crop height also changes during lodging, a preliminary analysis was performed to understand the variability of crop height. Over the same observation period, the crop height changed from a maximum of 159.3 cm–15 cm with a COV of 3.6% (Table 1) and was negatively correlated with CAI ($r = -0.55$) (not shown). The average crop height of a healthy crop was 79.32 cm while for VSL plots, it reduced to 36.15 cm. Crop height, however, did not correlate well with the remote sensing metrics. A

Table 3
The extracted metrics from RADARSAT-2 SAR data. * denotes the metrics referring to double bounce scattering, ** denotes the metrics referring to volume scattering, and *** denotes the metrics referring to surface scattering.

Method/Index	Parameters/Formulation	Reference
Backscattering coefficients	σ^o_{HH}, σ^o_{HV}, σ^o_{VH}, σ^o_{VV}	Lee and Pottier (2009)
Sinclair decomposition	$\lvert\sigma^o_{VV}\rvert^2$, $\lvert(\sigma^o_{HV} + \sigma^o_{VH})/2\rvert^2$, $\lvert\sigma^o_{HH}\rvert^2$	Krogager et al. (1997)
Pauli decomposition	Pauli_T11, Pauli_T22, Pauli_T33	Cloude and Pottier (1996)
Freeman-Durden decomposition	Freeman_double*, Freeman_volume**, Freeman_surface/odd***	Freeman and Durden (1998)
Yamaguchi decomposition	Yamaguchi_double*, Yamaguchi_volume**, Yamaguchi_surface/odd***, Yamaguchi_helix	Yamaguchi et al. (2005)
H-a Alpha decomposition	Entropy (H), Anisotropy (A), Alpha	Lee and Pottier (2009)
Cloude decomposition	Cloude_double*, Cloude_volume**, Cloude_surface/odd***	Cloude and Pottier (1996)
Touzi decomposition	Psi, Tau, Alpha, Phi	Touzi (2007)
Van Zyl decomposition	VZ_double*, VZ_volume**, VZ_surface/odd***	Van Zyl et al. (2011)
Coherency matrix elements	T11, T22, T33	Cloude and Pottier (1996)
Covariance matrix elements	C11, C22, C33	Cloude and Pottier (1996)
Span	$\lvert\sigma^o_{HH}\rvert^2 + 2\lvert\sigma^o_{HV}\rvert^2 + \lvert\sigma^o_{VV}\rvert^2$	Lee and Pottier (2009)
Pedestal height	$\frac{Maximum\ eigenvalue}{Minimum\ eigenvalue}$	Lee and Pottier (2009)
Radar vegetation index	$\frac{8\sigma^o_{HV}}{\sigma^o_{HH} + \sigma^o_{VV} + 2\sigma^o_{HV}}$	Kim and van Zyl (2009)
Radar forest degradation	$\frac{(\sigma^o_{HH} - \sigma^o_{HV})}{(\sigma^o_{HH} + \sigma^o_{HV})}$	Mitchard et al. (2012)
Canopy scattering index	$\frac{\sigma^o_{VV}}{\sigma^o_{VV} + \sigma^o_{HH}}$	Pope et al. (1994)
Biomass index	$\frac{(\sigma^o_{HH} + \sigma^o_{VV})}{2}$	Pope et al. (1994)
Volume scattering index	$\frac{\sigma^o_{HV}}{\sigma^o_{HV} + BMI}$	Pope et al. (1994)

Table 4
A post-hoc Tukey's HSD test is reported for different classes. * and ** indicate the statistical variances at 5% and 1% levels of significance.

S-1						
Class1	**Class2**	**VH**	**VV**	**Coh VH**	**Coh VV**	**VH/VV**
		P values				
H	**ML**	0.2400	0.0057**	0.9958	0.9955	0.6470
H	**SL**	0.0015*	0.0000**	0.9970	0.8467	0.6212
H	**VSL**	0.0000**	0.0000**	0.9980	0.9613	0.0641
ML	**SL**	0.9202	0.9661	0.9858	0.9907	0.9871
ML	**VSL**	0.0012**	0.1361	0.9877	0.9511	0.0393*
SL	**VSL**	0.0001**	0.0854	1.0000	0.6529	0.0069**

		RADARSAT-2 FQ8				RADARSAT-2 FQ21			
Class1	**Class2**	**HH**	**HV**	**VH**	**VV**	**HH**	**HV**	**VH**	**VV**
		P values				*P values*			
H	**ML**	0.8336	0.2815	0.2505	0.4107	0.7598	0.9892	0.9821	0.5832
H	**SL**	0.7432	0.0059**	0.0058**	0.0118*	0.8011	0.1838	0.1481	0.0048**
H	**VSL**	0.0000**	0.0000**	0.0000**	0.0000**	0.7661	0.0000**	0.0000**	0.8146
ML	**SL**	0.9966	0.9896	0.9947	0.9774	0.9923	0.6921	0.6793	0.6198
ML	**VSL**	0.0110*	0.0080**	0.0070**	0.3200	0.9953	0.0002**	0.0003**	0.3064
SL	**VSL**	0.0000**	0.0001**	0.0001**	0.1818	0.9999	0.0004**	0.0006**	0.0024**

Table 5
Pearson correlation coefficients (r) between CAI and the metrics derived from S-1 data.

	Satellite metrics	r
1	VH	0.67
2	VV	0.48
3	Coh VH	−0.31
4	Coh VV	−0.34
5	VH/VV	0.25

detailed time series analysis of S-1 derived metrics was also carried out for different lodging stages to understand the effect the rainfall, wind speed and other crop parameters (such as LAI) (not shown).

3.1. Scattering characteristics of different lodging stages

The scattering characteristics of different lodging stages from the metrics derived from S-1 and RADARSAT-2 multi-incidence angle data are presented in Fig. 7. Only results related to the variation of backscatter and coherence metrics are discussed in this section. The Pearson correlation coefficients of the other metrics are presented in Table 6.

Significant differences in the mean values of the metrics were found among the four lodging stage groups. It can be seen in Fig. 7a that the σ^o_{VV} and σ^o_{VH} derived from S-1 increased as lodging became severe. The increase was more prominent in VH polarization with a difference of 1.3 dB, 2 dB and 4.2 dB for ML, SL and VSL stages respectively w.r.t. H stage. The σ^o_{VV} variation was smaller (2.7 dB difference between H and VSL) (Fig. 7a). The average σ^o_{VH} and σ^o_{VV} for H stage was around −19 dB and −12.5 dB respectively, whereas for VSL, it was around −14.8 dB and −9.8 dB, respectively (Fig. 7a). The change in S-1 derived $\sigma^o_{VH/VV}$ was very gradual while no significant variations were observed for RADARSAT-2 data (not shown). From Fig. 7a, the average $\sigma^o_{VH/VV}$ for H, ML and SL stages is almost the same while for VSL stage, it increased by 1.4 dB (w.r.t. H). From Fig. 7b, it can be observed that coherence in VV drops substantially when lodging occurs. This is also evident from Table 5, where CAI had a low negative correlation with VV ($r = -0.34$) and VH ($r = -0.31$) coherence. However, the variation in coherence was statistically insignificant across different groups (Table 4). This may have been due to the unavailability of field observations every six days. Thus, in our case study coherence did not serve as a reliable discriminator between different lodging stages.

In the case of RADARSAT-2 data, the scattering mechanisms were investigated at low/steep (FQ8) and high/shallow (FQ21) incidence angles by interpreting the SAR data behaviour at C-band for different

Table 6
Pearson correlation coefficients (r) between CAI and the metrics derived from RADARSAT-2 FQ8 and FQ21 data. * denotes the metrics referring to double bounce scattering, ** denotes the metrics referring to volume scattering and *** denotes the metrics referring to surface scattering. The backscattering coefficients are expressed in dB and the covariance/coherency matrix elements are in the linear scale.

	Satellite metrics	r Radarsat-2 FQ8	r Radarsat-2 FQ21
1	HH	0.19	−0.42
2	HV	0.74	0.40
3	VV	0.48	−0.27
4	Span	0.30	−0.23
5	Pedestal height	0.57	0.21
6	RVI	0.55	−0.11
7	RFDI	−0.49	−0.24
9	CSI	0.22	0.30
9	VSI	0.48	0.17
10	BMI	0.17	−0.21
11	Sinclair_1	0.41	−0.0202
12	Sinclair_2	0.65	−0.005
13	Sinclair_3	0.15	−0.37
14	Pauli_T11	0.30	−0.33
15	Pauli_T22	0.65	−0.006
16	Pauli_T33	0.24	−0.15
17	FD_dbl*	−0.40	−0.42
18	FD_vol**	0.67	0.00
19	FD_surf***	−0.54	−0.15
20	Yamaguchi_dbl*	−0.40	−0.53
21	Yamaguchi_vol**	0.69	0.05
22	Yamaguchi_surf***	−0.45	−0.26
23	Yamaguchi_hlx	−0.20	−0.08
24	VZ_dbl*	−0.45	−0.53
25	VZ_vol**	0.66	0.03
26	VZ_sur***	−0.11	−0.08
27	Cloude_dbl*	−0.05	−0.43
28	Cloude_vol**	0.62	−0.07
29	Cloude_surf***	0.22	0.23
30	Entropy	0.52	0.26
31	Anisotropy	−0.33	−0.39
32	Alpha	0.34	0.18
33	Psi	0.18	−0.01
34	Tau	0.12	−0.09
35	Alpha_touzi	0.11	0.07
36	phi	−0.26	−0.25
37	T11	0.27	−0.14
38	T22	0.35	−0.30
39	T33	0.72	0.02
40	C11	0.24	−0.20
41	C22	0.42	−0.16
42	C33	0.44	−0.06

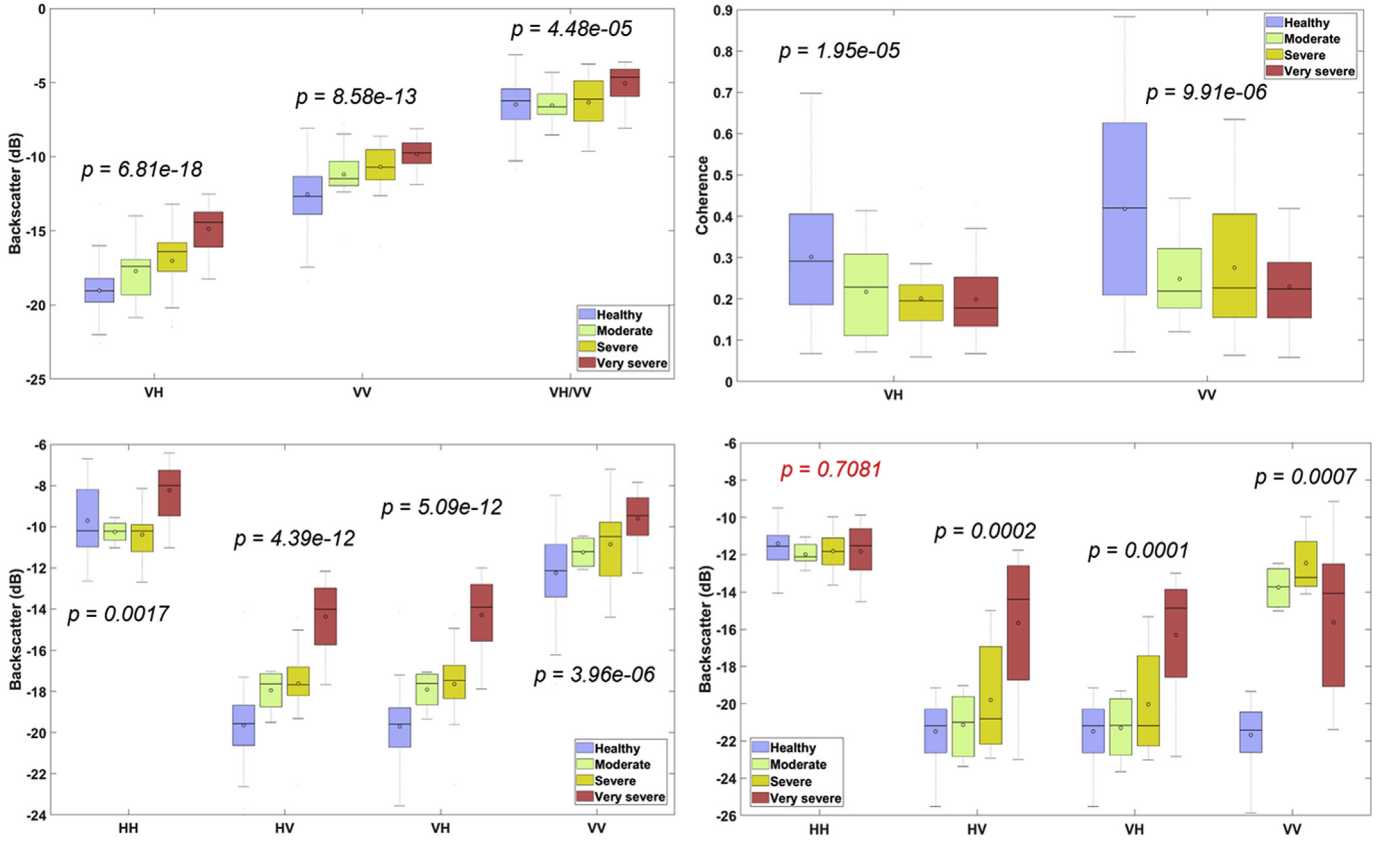


Fig. 7. Box plots of (a) S-1 VH, VV, VH/VV σ^o (n = H:50, ML:11, SL:28, VSL:29), (b) S-1 VH and VV coherence (n = H:50, ML:11, SL:28, VSL:29), (c) RADARSAT-2 FQ8 HH, HV, VH, VV σ^o (n = H:22, ML:4, SL:15, VSL:16) and (d) RADARSAT-2 FQ21 HH, HV, VH, VV σ^o (n = H:29, ML:6, SL:13, VSL:13). The p-values calculated using the Kruskal Wallis test determines the significance of the difference in sample means between different lodging stages (at 0.05 level of significance). The red p-values are not significant at $p = 0.05$ level of significance. (*2-column fitting*). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

lodging stages. Fig. 7c and d compare the σ^o_{HH}, σ^o_{HV}, σ^o_{VH}, and σ^o_{VV} values observed at approximately 27° and 41° over different stages of lodging in wheat. The average σ^o_{HV} and σ^o_{VH} was almost the same since natural targets (e.g. agricultural fields) follow the assumption of reciprocity (Larranaga et al., 2012; Lee and Pottier, 2009). σ^o_{VH} was thus not included in the subsequent analysis. In general, studies show that σ^o decreases over a crop field with moderate surface roughness as the angle of incidence increases (McNairn and Brisco, 2004). This is noticeable in the H stage, as the incidence angle increased from 27 to 41°, the overall σ^o decreased by ~0.6 dB. The σ^o measurements showed distinct behaviours at the two incidence angles (Fig. 7c and d). For instance, the average σ^o_{HH} moderately increased (by 1.5 dB) as CAI increased at 27° while it decreased steadily at 41° (1.6 dB). Remarkable differences were also noticeable in the HV band. At low incidence angle (Fig. 7c), a sharp increase was observed in σ^o_{HV} (5.3 dB) as the stage changed from H to VSL, whereas at higher angles, a relatively small increase (1.4 dB) from H to ML stage was observed, after which the σ^o_{HV} saturated.

While statistically significant differences in the Kruskal Wallis test were found between the H, ML, SL and VSL classes for most of the metrics, the post hoc Tukey's test showed significant differences only between some of them (Table 4). For instance, in the case of S-1 σ^o_{VV}, the H class was significantly different from the lodging classes (ML, SL and VSL) but the separability among the lodging classes was not significant (Table 4). On the other hand, significant differences were observed among the lodging classes (ML and VSL, SL and VSL) with σ^o_{VH} (Table 4). With RADARSAT-2 data, the results confirmed that the σ^o_{HV} in FQ8 beam mode was favourable for distinguishing between most of the classes (H-SL, H-VSL, ML-VSL, SL-VSL) while σ^o_{HH} (H-VSL) and σ^o_{VV} (H-SL, H-VSL) resulted in reasonable detection of some of them. The results are presented in Table 4.

3.2. CAI estimation using support vector regression

A preliminary test, employing a neighbourhood component analysis (NCA) feature selection method, was first conducted to compare the performance of SVR for two cases: the model with the whole set of RS metrics and the one with subset features. The results were quite similar, with lower RMSE in the former case, which confirmed that multicollinearity was dealt within the SVM and the complementary effect of the interaction among the predictor variables contributed to increased accuracy of the model (results not shown). Thus, the whole set of features were used in the final regression model.

As shown in Fig. 8, SVR models were evaluated for estimating CAI from satellite-derived metrics with inputs from (a) S-1, (b) RADARSAT-2 FQ8 (FQ8 model) and (c) RADARSAT-2 FQ21 (FQ21 model). The models were calibrated, cross-validated and the overall accuracy and error statistics were calculated. The three SVR models with Gaussian radial basis functions were trained for CAI values ranging from 1° to 80° using the 47 SAR metrics available for the different sensors (see section 2.4 for details).

The utility of the S-1 derived metrics was confirmed by a good linear relationship that was established throughout the growth cycle (R^2_{CV} = 0.78; $RMSE_{CV}$ = 11.63°) (Fig. 8a), with only a minor underestimation at higher CAI values (> 70°). The model performance considerably improved with the RADARSAT-1 FQ8 model. The FQ8 model performance was most robust since it explained about 87% of the observed variance in CAI and the $RMSE_{CV}$ dropped by almost 24% in comparison to S-1 model (Fig. 8b). However, a slight underestimation

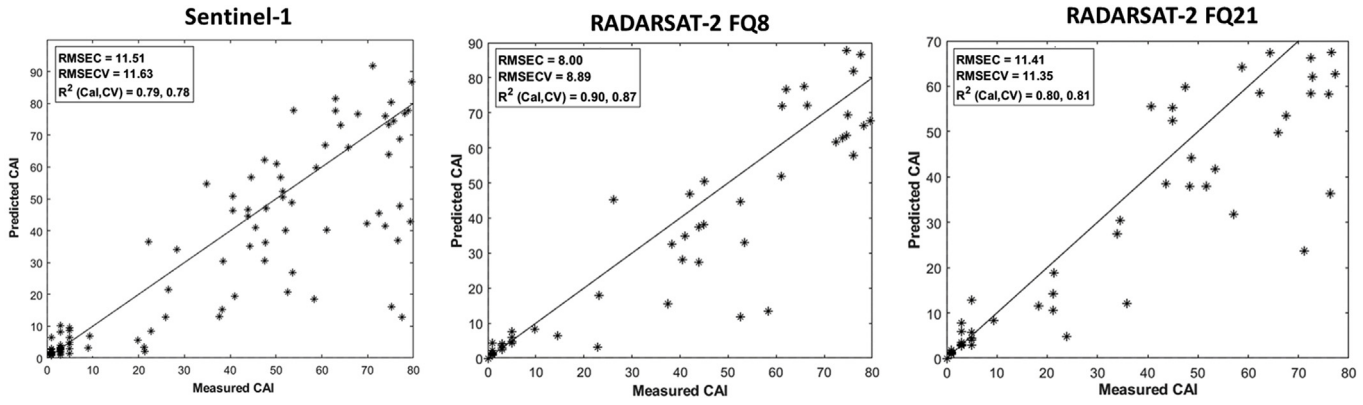


Fig. 8. Predicted versus measured crop angle of inclination (CAI) from S-1 and RADARSAT-2 data using support vector regression. (a) S-1: model calibration and cross-validation ($n = 118$), (b) FQ8: model calibration and cross-validation ($n = 57$), (c) FQ21: model calibration and cross-validation ($n = 61$). (*2-column fitting*).

was again recorded in the FQ8 model, which was most evident at medium CAI (50-60°) (see Fig. 8c). The performance of the FQ21 model, was comparable to the S1 model as it explained close to 81% of the variance in the plot level measurements, with some underestimated values at high CAI (> 70°). In contrast to the FQ8 estimations, the $RMSE_{CV}$ increased by 28%.

4. Discussion

This study investigated the performance of S-1 and RADARSAT-2 data for estimating CAI in wheat. We first assessed the scattering behaviour of metrics derived from SAR satellite data across different lodging stages. Further, we established quantitative relationships between CAI measured in the field and RS-based metrics. Finally, we used those relationships to map CAI in 26 wheat fields in the study site. The important findings are discussed in this section.

4.1. Scattering characteristics of SAR metrics

The wavelength of S-1 and RADARSAT-2 satellites (~5.5 cm) is comparable to the size of some of the plant components such as grain heads and leaves of the wheat crop, which makes them particularly useful for monitoring wheat crop (Ferrazzoli et al., 1997). In section 3.1, we saw how the backscattering coefficients (and coherence from S-1) derived from the two sensors with different configurations vary across different lodging stages. The relative change in S-1 σ^o_{VV} was smaller than that in σ^o_{VH} as lodging became severe (Fig. 7a) which is due to increased attenuation in VV. The high values of σ^o_{VH} indicate that volume scattering from the crop has a dominant contribution, especially at high incidence angle (40°). Studies show that σ^o_{VH} and σ^o_{VV} are also sensitive to vegetation-soil interactions (double bounce effect) (Picard et al., 2003). When the crop is lodged, the vegetation-soil double scattering increases with an increase in CAI, which explains the increase in σ^o_{VH} and σ^o_{VV} polarizations. The Pearson coefficient values in Table 5 also confirm our interpretation that σ^o_{VV} and σ^o_{VH} increase significantly with CAI (positive r). $\sigma^o_{VH/VV}$, on the other hand, was particularly sensitive to the VSL stage (Fig. 7a) as a consequence of high sensitivity of σ^o_{VH} w.r.t. σ^o_{VV}. The $\sigma^o_{VH/VV}$ is known to be strongly correlated with crop height and represents the randomness of scattering mechanisms (Canisius et al., 2018). During lodging, the crop height reduces with an associated increase in CAI. The reduction in crop height is more drastic in the VSL stage (mean height 36.73 cm), and it seems that $\sigma^o_{VH/VV}$ is sensitive to this change.

Furthermore, no meaningful differences were observed between the coherence of different lodging stages as the change in coherence values was statistically insignificant (Table 4). Overall with a six-day temporal baseline, the coherence was low over the vegetated fields. However, an inverse relationship was found between CAI and coherence (VV and VH, see Fig. 7b). Observed S-1 coherences of wheat declined as the CAI increased (Fig. 7b). A probable explanation for this is that a lodged crop screens the ground more effectively, resulting in greater backscatter from vegetation that decorrelates more quickly than that from the underlying soil (Engdahl et al., 2001). In a simplistic manner, coherence can be explained by the amount of incoherent and coherent scattering from the soil surface as well as the vegetation (Blaes and Defourny, 2003). In the case of a healthy crop, it is possible that mainly coherent scattering from the soil constitutes the backscattering and results in higher coherence values than that from a lodged crop. However, due to a limited amount of data (every six days) and lack of literature (for six-day interval), it is difficult to comment on how the underlying soil moisture and surface roughness affect the soil scattering in healthy and lodged areas in this study. Precipitation also results in temporal decorrelation, thus complicating the interpretation of coherence (Tamm et al., 2016). In this study we observed that in cases where rainfall coincided with the date of image acquisition (e.g. 13 May 2018–2.6 mm and DoY 153 i.e. 6 June 2018–7.6 mm) (figure not shown), it may have resulted in increased soil moisture. In addition, the antecedent soil moisture conditions on other dates (due to few other rainfall events) could also have resulted in a decrease in the coherence values, thus impeding the detection of a lodged event. Therefore, repeated observations across several growing seasons are necessary to confirm the potential exploitation of coherence as a key source of information for lodging assessment.

Some interesting findings were also generated from high (41°) and low incidence angle (27°) RADARSAT-2 data. High incidence angle is known to maximize the contribution of vegetation scattering due to increased path length (of the signal) through the crop canopy, whereas low incidence angle is less sensitive to vegetation attenuation and maximize the contribution from ground scattering in the return signal (Srivastava et al., 2009). At both angles, the σ^o_{HH} is higher than that in σ^o_{VV} (Fig. 7c and d). This finding is consistent with the findings of Brown et al. (2003) and Mattia et al. (2003) for H class, who found that σ^o_{HH} measurements for healthy wheat are higher than σ^o_{VV}. This is primarily due to stronger attenuation of the *V* wave by the vertical stems on both forward and return propagation paths (Xu et al., 2014). The angular variation was particularly evident between H and VSL stages (Fig. 7c and d). At low incidence angle, the path length of a radar signal through vegetation is minimized (Brown et al., 2003). It is possible that as CAI increases, this path length through the lodged crop is further reduced, resulting in higher scattering from the ground. Given the simplicity of our regression model, it cannot account for the effect of soil on the total backscatter.

Since soil return dominates the σ^o_{HH} (Mattia et al., 2003), the σ^o_{HH} increases as lodging become severe. At low incidence angle (Fig. 7c), a sharp increase in σ^o_{HV} as the stage changes from H to VSL stage (5.3 dB) is due to contributions from both volume and double-bounce scattering.

At low incidence angle, σ^o_{HV} has a good correlation with CAI (r = 0.74) while at high incidence angle, a moderate correlation exists (r = 0.40) (Table 6). The ability of σ^o_{HV} to distinguish most of the lodging stages at low incidence angle (Table 4) can be explained by the strong impact of ear bending/ear orientation on σ^o_{HV} (Ferrazzoli, 2002). The Pearson coefficient values in Table 6 also confirm our interpretation that volume scattering indicators increase significantly with CAI (positive r) while the double bounce indicators show a negative trend (negative r).

Lodging mainly results in the irregular appearance of crop canopy and changes the orientation of the canopy elements. In a healthy plot, the crop canopy stands erect with the horizontal orientation of the ears. When lodging occurs, the stems and the ears bend downwards and incline against each other. This effect is more pronounced in VSL stage. Radar σ^o is sensitive to these structural changes and hence can explain the variability in different stages of lodging (Fig. 7a–d). The variation is larger at the low incidence angle (Fig. 7c). Thus, the box plots in Fig. 7 provide evidence for the potential application of remote sensing (or specifically C-band) to discriminate different stages of lodging. We recommend that future studies explore simple electromagnetic models to better understand the behaviour of different polarizations in the event of lodging.

4.2. Estimation of CAI

Satisfactory results were achieved when we modelled the field measured CAI values using SVR. Overall, the SVR models fitted with inputs from RADARSAT-2 data performed better than the S-1 model. Microwave scattering from vegetation is dependent on the SAR frequency/wavelength, incidence angle and polarization (Soria-Ruiz et al., 2009). Apart from this, the spatial resolution, radiometric quality and date/time of data acquisition also affect the backscatter response (Bovenga et al., 2018) and may provide different and/but complementary information. Therefore, even though these two sensors have the same frequency, some of their characteristics such as polarization, incidence angle (for FQ8), radiometric quality, spatial resolution and acquisition date/time are considerably different resulting in a better performance for RADARSAT-2. Among the input variables from RADARSAT-2, the metrics referring to volume scattering derived from low incidence angle data (FQ8) were highly correlated to CAI ($0.62 < r < 0.69$) while those referring to double-bounce scattering had a stronger negative relationship with CAI at a high angle of incidence (FQ21) ($-0.30 < r < -0.53$) (see Table 6). Pedestal height (PH) and RVI also proved to be important indicators of CAI at a low incidence angle ($r = 0.57, 0.55$) (Table 6). The height of the pedestal determines the degree of polarization of the scattered wave. The signatures with high PH are characteristic of targets that are dominated by volume scattering (McNairn et al., 2002). PH is also reported to be directly proportional to vegetation density (Evans et al., 1988), which increases with the increase in lodging percentage (Sher et al., 2018). RVI (ranges from 0 to 1), on the other hand, is a measure of randomness of scattering. The average RVI for the healthy vegetation was close to 0.6 while for the VSL crop, RVI was considerably higher (> 0.96) at a low incidence angle (see Table 6). Due to the displacement of some of the individual plant scatterers during lodging, several scattering mechanisms arise from the target resulting in an increased degree of random scattering.

The findings also revealed that low incidence angle data is particularly sensitive to high CAI (> 60°) while high incidence angle can be useful for predicting lower CAI values (Fig. 8b and c). The underestimation of the FQ8 model for the case of SL could be due to the non-homogeneous distribution of the number of samples in different lodging classes. It is possible that the SVR model did not have enough cases in the SL class to be more robust. In addition, in our case, the samples for the SL class comes from two images while those for VSL come from one image which could also have been a potential source of error. The reason being that during cross-validation, the model has to estimate VSL on the same target condition while SL can correspond to different target conditions according to the acquisition dates. For S-1 data, σ^o_{VH} and σ^o_{VV} had a reasonable correlation with CAI ($r = 0.67$ and $r = 0.48$, respectively) while coherence values were poorly correlated with CAI ($r < -0.4$) (Table 5). The performance of the S-1 model was almost comparable to FQ21 model, which suggests that the potential of S-1 data; considering the significance of free data for operational needs, cannot be ignored.

It is important to note here that the acquisition of S-1 and RADARSAT-2 FQ8 data in the ascending pass (evening time) (Table 2) ensured that the possible early morning dew effects on the backscatter values were non-existent. Due to user conflicts and acquisition constraints, the RADARSAT-2 FQ21 images were however, acquired in the descending pass (morning 5 a.m.). It is possible that the morning dew could be a source of error in the analysis of these images. However, a study done by Wood et al. (2002) suggests that there is a significant correlation between the backscatter of ascending and descending orbits, which implies that although absolute backscatter increases in the presence of dew, relative differences remain very similar. In addition, the effect of azimuth angle or orientation on the polarimetric response of wheat crop could be neglected in this study since for C-band (contrary to lower frequencies such as L-band), the scattering value is independent of azimuth angle as shown by Stiles et al. (2000).

4.3. Mapping of CAI

To map CAI, the SVR models were applied to two S-1 and two RADARSAT-2 images. The non-wheat areas were masked out, and four maps of predicted CAI were generated. The resulting maps are shown in Fig. 9. The maps were derived when the crop was in the milk/dough development stage (31 May), early ripening stage (6 June) and mid-ripening stage (13 June). The healthy areas are mapped with CAI values close to 0° while the most severe ones are close to 90°. The maps indicate that lodging was widespread across the wheat fields in the study area with more severely lodged areas detected in the ripening stage as expected. However, since field observations were used to assess the mapping results qualitatively, the obtained CAI maps should be considered as qualitative indicators of the within-field spatiotemporal variability of lodging. However, patterns provided by the different models show the convergence of CAI results, supporting the general qualitative validity of the produced maps. These maps can have potential applications for crop management and precision agriculture. For instance, the CAI maps generated before harvesting can contribute to in-field navigation routes to guide drivers or autonomous driving vehicles to adjust their speeds based on the lodging condition and thus minimize the harvesting losses.

The closest acquisition dates between S-1 and RADARSAT-2 were taken into account for the comparison. For instance, Fig. 9a and c shows the S-1 and RADARSAT-2 FQ21 predicted CAI maps for 31st May 2018. A closer look at the two figures demonstrates that in FQ21, variability in CAI values is more effectively captured. This is particularly evident in field 4 of Fig. 9c. While the S1 model mapped all of field 4 as H (Fig. 9a), the FQ21 model could capture the ML areas as well as expected (Fig. 9c). Also in field 1 (Fig. 9c), the S1 model overestimated the healthy patches. The slightly more precise estimation of CAI from the FQ21 model can be attributed to its high spatial resolution that allowed for the assessment of spatial variability at both field and plot scale. The FQ8 image from 13th June 2018 could also capture the spatial variabilities of CAI quite effectively. These spatial maps can serve as a valuable baseline for assessing the performance of the models across stages of lodging progression.

5. Conclusions

In this study, we introduced a new method to assess crop lodging stages based on the crop angle of inclination (CAI) derived from S-1 and

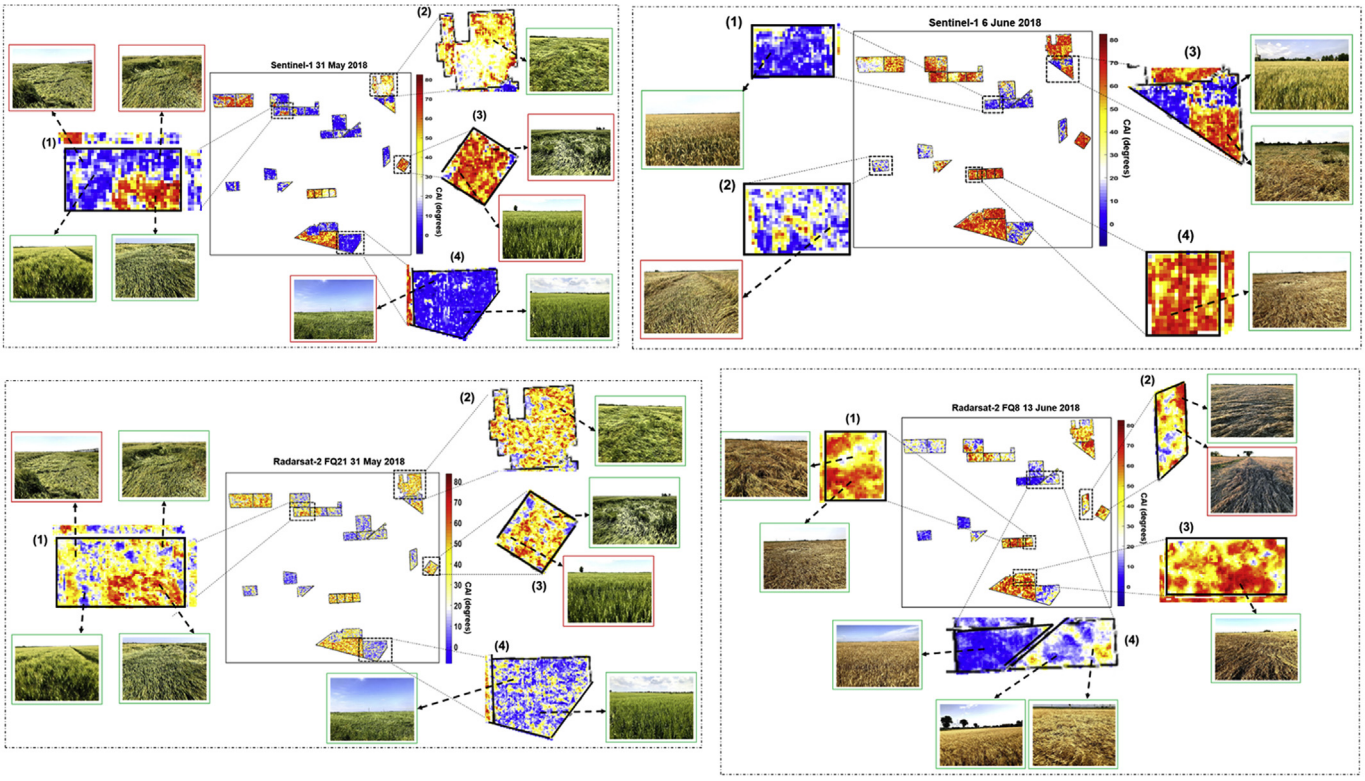


Fig. 9. Multi-temporal CAI maps predicted from (a) S-1 data acquired on 31st May 2018, (b) S-1 data acquired on 6th June 2018, (c) RADARSAT-2 FQ21 data acquired on 31st May 2018, and (d) RADARSAT-2 FQ8 data acquired on 13th June 2018. Field observations (photographs) from a number of plots have been used to examine the maps. The correctly mapped areas are supported with field photographs marked green while the others are marked red. "RADARSAT-2 Data and Products. MacDonald, Dettwiler and Associates Ltd. (2018) – All Rights Reserved. RADARSAT is an official trademark of the Canadian Space Agency." (*2-column fitting*). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

RADARSAT-2 data. We first demonstrated the potential of metrics derived from these datasets to discriminate between different lodging stages: non-lodged/healthy (H), moderate lodging (ML), severe lodging (SL) and very severe lodging (VSL). The performance of different support vector regression models (SVR) was then evaluated for estimating CAI from SAR satellite-derived metrics. The main conclusions of the study are summarized below:

1) Among the different SVR models that were developed to establish quantitative relationships between field-measured CAI values and RS metrics, the RADARSAT-2 FQ8 (low incidence angle) performed most robustly with an R^2_{CV} of 0.87 and $RMSE_{CV}$ of 8.89°. The performance of S-1 and RADARSAT-2 FQ21 (high incidence angle) models were comparable with an $RMSE_{CV}$ of 11.35° and 11.63°, respectively. The differences in sensor configuration (mainly the incidence angle and spatial resolution) and temporal data availability had a primary influence on the model performance while the increased dimensionality of RADARSAT-2 data played a secondary role. However, the combination of the two can improve the ability to monitor CAI throughout the crop growth cycle.
2) Low incidence angle data was found to be particularly sensitive to high CAI (> 60°) while high incidence angle predicted the lower CAI values more accurately.
3) Among the polarimetric decomposition parameters that were derived from RADARSAT-2 data, the volume scattering parameters (such as FD_vol, Yamaguchi_vol, VZ_vol, Cloude_vol) derived from low incidence angle data (FQ8) were highly correlated with CAI ($0.62 < r < 0.69$) while double-bounce scattering parameters (such as FD_dbl, Yamaguchi_dbl, VZ_dbl, Cloude_dbl) had a stronger negative relationship with CAI at a high angle of incidence (FQ21) ($-0.30 < r < -0.53$).
4) Dense time series of high resolution data provided by S-1 can be exploited for lodging stage assessment. The S-1 model explained 78% of the CAI variability in the area and the unprecedented amount of free S-1 data guaranteed with the next generation of Sentinel up to or probably beyond 2030, presents a unique opportunity to monitor lodging in crops in almost real time.
5) In summary, this study provides evidence of the potential of high-resolution SAR remote sensing data in estimating CAI as a measure of lodging severity assessment, which to the best of our knowledge, has not been documented in the literature.

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